Supplementary Evidence: When Do LLM Side-Channel Defenses Protect Agent Workflows?

October 21, 2026

This supplement provides the search accounting, corpus maps, saved empirical results, dual-coding records and source-bound audits accompanying the main paper. The package README maps these sections to machine-readable data and the read-only verifier. Original failed and negative outcomes are preserved alongside positive results.

S1 Search and Evidence Audit

TABLE S1
Historical route accounting for the original October 8, 2026 snapshot, including its later source checks. Counts overlap and do not form a complete PRISMA flow; the October 21 supplementary round is reported separately.

Route / disposition	Count
Broad IEEE Q4 hits; not fully screened	35,743
Amended Q4b export; recall not verified	847
IEEE saved export occurrences / distinct records	2,079 / 2,001
IEEE full-text referrals / background routes	25 / 205
ACM saved occurrences / publication groups	16,687 / 13,201
ACM Oct. 14 provisional exclusions / full-text referrals / unresolved	9,368 / 883 / 2,950
ACM dispute groups / combined check candidates (Oct. 18)	332 / 57
57-item source check: scoped dispositions / unresolved	57 / 0
Backward reference occurrences (40 fixed seeds)	2,237
Forward seeds resolved / distinct returned works	38 / 24
Recent finite reconciliation: overlapping decision rows	109
Source-limited candidates not admitted / background chapter	4 / 1
Snapshot core / adjacent studies	43 / 17
Snapshot coded study families	60

The 57 scoped dispositions are not completed full readings. B069's complete official abstract supports background routing; later-page eligibility remains unchecked. The 109 rows overlap; 883/2,950 are dated global routes. Q4b does not replace unscreened Q4. This snapshot's coded-set cutoff is October 8, 2026; its counts do not describe the later expanded collection.

Targeted discovery and seed-anchored snowballing Because native IEEE/ACM keyword exports (Table S2) captured only 1/24 and 0/24 of the 24 core seed papers, our primary acquisition combined venue-targeted proceedings screening (USENIX Security, NDSS, IEEE S&P, CCS, PMLR), targeted arXiv discovery, and one-hop backward (2,237 reference occurrences) and forward (24 distinct works across 38/40 resolved seeds) citation tracing from 40 fixed seeds, followed by the completed 565-record October 21 Semantic Scholar and native arXiv search frame (Section S10).

Historical frame accounting and independent dual-coding Every prepared metadata ID in the historical frame has an explicit disposition (4,753 unique IDs: 2,778 exclusions, 116 referrals, 1,859 unresolved routes; Table S1). While historical exclusion checks (1,300 checks in the 1,555-record frame; 23-case AI-on-AI exercise with 17/21 determinate agreement; 332-dispute readbacks) used the single AI-assisted workflow, the revised package supplements them with independent human/expert dual-coding across 100 stratified screening records ($\kappa = 0.886$), 144 split-audit cells ($\kappa = 0.876$), 24 retrospective transfer cases ($\kappa = 0.866$) and 150 pairwise comparability judgments ($\kappa = 0.916$; `experiments/revision-m1-m8/`).

S2 Family Index, Corpus Maps and Prior Work

The accompanying *Supplementary Evidence* archive is the access route for the full tables and fit-free checks. Its README starts with the current paper, family index, result ledger, utility/cost inventory and claim links. Each result joins a citation-key family, source version/hash and locator to its receiver/intervention fields and supported judgment; Table S3 is a filled example. Separate case-evidence entries identify additional source subconditions, rather than silently pooling them into the 33 scoped inventory records.

The merged index enumerates 82 families: 58 core (20 agent-native, 38 runtime/application-inherited) and 24 adjacent, including five authorized diagnostic components. The original 60-family index is a separately labeled historical snapshot. Mixed semantic attribution remains adjacent under the auxiliary-only rule. The package preserves the snapshot's 33 scoped result records/66 field positions and the complete original empirical and workflow analyses. S10 supplies the 22 added-family rows and expansion sensitivity. Source locators and hashes support inspection; third-party full texts/raw traces are obtained from their named sources.

Roles of the remaining indexed sources Additional core studies concern content/hidden-property inference [1, 2, 3, 4, 5, 6], identity/provenance attribution [7, 8, 9, 10], and authorized integrity diagnostics [11, 12, 13]. Adjacent audit, execution, browser-data and profiling studies supply boundary comparisons rather than additional auxiliary privacy attacks [14, 15, 16, 17, 18, 19, 20, 21]. The index records each receiver, target relation and evidence scope; the reference list includes these

TABLE S2
Executed native queries and saved export occurrences. IEEE dates: October 8, 2026; ACM query/partition acquisition: October 8–9, 2026. Counts are not included study families.

ID	Exact query	IEEE saved/hits	ACM saved/original hits
Q1	`("large language model" OR LLM) AND ("side channel" OR "side-channel" OR "timing attack")`	63/63	479/479
Q2	`(agent OR agentic) AND ("traffic analysis" OR fingerprinting OR "metadata leakage" OR "side channel" OR "side-channel")`	671/671	6,181/6,181
Q3	`("KV cache" OR "prompt caching" OR "speculative decoding") AND (privacy OR leakage OR attack)`	37/37	511/511
Q4	`("browser agent" OR "research agent" OR "tool calling" OR "model context protocol" OR "multi-agent") AND (timing OR fingerprinting OR leakage)`	0/35,743	Not run
Q4b	`(("browser agent" OR "research agent" OR "tool calling" OR "model context protocol" OR "multi-agent") AND (timing OR fingerprinting OR leakage) AND ("large language model" OR LLM))`	847/847	4,026/4,026
Q5	`("language model" OR LLM) AND ("traffic shaping" OR "cache isolation" OR padding OR "side-channel defense")`	95/95	3,689/3,688
Q6	`("language model" OR chatbot OR "AI assistant" OR "generative AI") AND ("side channel" OR "side-channel" OR "traffic analysis" OR fingerprinting OR "timing leakage")`	366/366	1,801/1,801

IEEE used Command Search/default metadata, without a lower-year or document-type filter; every result page was exported as RIS with Citation and Abstract. Q4's zero means not exported in the October 8 acquisition, not zero hits or a completed screen. Q4b supplements Q4; recall preservation is unverified. The later Q4 prefix does not reconstruct historical membership. ACM used the All field in the ACM Full-Text Collection (not Guide), then native E-Publication Date partitions and All Results/BibTeX export. Partition receipts retain exact dates and URLs. Q4b's June bin has 285 displayed versus 284 saved; agreement of the overall saved/original counts does not repair the missing identity. Q5's partition total differs from its original whole-query count. Final-named export files were byte/hash matched on October 9; these two retrieval discrepancies remain. The October 8 coded-set cutoff is checked separately from acquisition partitions.

TABLE S3
Filled record and its licensed judgment. Values are source-reported; no cross-mode effect is computed.

Join key / field	Entry
Family / result	`dangol2026workflow` / `DANGOL-livebinding`.
Version / locator	arXiv:2606.07150v3, §9.7, live-protocol paragraph after Table 5 [22].
Target / view	Workflow classification from live Oblivious HTTP (OHTTP)/Agent-to-Agent (A2A) traffic; not GUI-state observation.
Intervention	Live fixed-rate cover with fresh per-session identifiers, distinct from transformed stored captures.
Workload / testbed	Workflow classes with client, relay, gateway and agent co-located on one loopback host.
Outcome / cost	Accuracy 0.30; about 13× bytes and 11× latency in that condition.
Unresolved fields	Objective terminal task utility NR-R; precise paired live-refit comparison NR-V.
Licensed judgment	These costs describe this live protocol condition. They neither price dummy GUI actions nor prove task preservation. A matched objective endpoint is needed for the latter.

indexed sources, not just the central case papers.

S2.1 RQ sensitivity

The original expansion adds procedural skills, physical receivers, topology shifts and architecture timing to RQ1–2. The latest finite search additionally changes the positive defense evidence: CachePrune and OTRO supply useful mechanism-bound privacy/quality/performance conditions. RQ3 still requires generator-specific bridges; RQ4 separates simulation, component overhead, fixed guardrails and executed protection. S10 reports these changes explicitly. Section S6 retains the original source-deletion analysis; the new controlled experiments independently instantiate its first two placement mechanisms.

S3 RETAINED FAILURE-INCLUSIVE CONTROLS

The 288-trace fixed-policy pilot has call-count-determined labels and saturated coarse F1=1.00, so it supports no sequence-increment claim. In the separate tool-front-end study, the observer sees both tools; the constrained baseline succeeds on 19/24 tasks, below the frozen promotion gate. These local results are not passive-network or deployed-defense evidence.

The retail case retains 148 windows, 49 empty observations and 102 stopped-error episodes. All three frozen loss contrasts have intervals including zero; test goal achievement is 7/40 versus 8/40 and joint utility remains unresolved. Neither this null-compatible estimate nor failed tasks establish absence of leakage or effective defense. Full protocols and results remain in the delivered evidence package.

S4 PUBLIC-TRACE REANALYSIS: PROTOCOL AND BOUNDARIES

UI Traces (commit `4f3df18`) supplies 4,200 records/14 declared identities, retaining 1,806 error/absent-result indicators. Labels are declarations, not authenticated backends or task correctness. The supplemental package binds source bytes, code and receipts.

Predictors use JS types, local time and callback order, excluding collector time/navigation, URL/DOM text, questions, answers and diagnostics. Cross-document order assumes an archive receiver, not authenticated website capture. Exact-question-disjoint train/validation/test groups have 2,043/1,022/1,135 rows and 395/202/226 questions (823 distinct questions). The source split identifies FRAMES but has no semantic task-family field. Full identity–dataset/exact-question support and a seven-bin first-word *grammatical proxy*, explicitly not a semantic family, are supplemental; neither semantic

TABLE S4
Primary observable/receiver profile × signal origin across all 58 core families (showing Snapshot 43 → Full 58). Column n counts distinct families; each family occupies one primary receiver profile row but may span multiple generating origins (P1–P8).

Observable / receiving permission	n (43→58)	P1	P2	P4	P5	P6	P7	P8	Representative sources
Flow / destination / routing metadata	12 → 16	0	10	1	7	2	2	3	AgentPrint; WRA; NetEcho; LLM-WFIN; Routing Metadata
Own-site actions / browser / harness	5 → 8	0	0	1	8	2	6	0	UI Traces; FP-Agent; WebAgents;AgentProv; Lidar
API duration / cache / split-return	12 → 14	0	14	1	0	0	0	0	Early Bird; LeakyLMs; CachePrune; SplitDecoy
Resource / physical / sparsity / page	10 → 15	15	6	0	1	0	0	0	TDXRay; OTRO; PAGE404; SparLeak; SparSEEty; PhysicalEdge
Trajectory / procedure / diagnostic log	4 → 5	0	1	1	5	2	0	0	SIGLEAK; ClawGuard; cycle detection; MIRAGE (diagnostics)
Total distinct core families	43 → **58**	Each family contributes once to n (20 workflow-origin A, 38 inherited-runtime/application B).							

P1 hardware; P2 serving; P4 application/API; P5 orchestration; P6 tools; P7 environment; P8 multi-agent. P3 denotes transport (coded as a receiving projection). Five Type IV authorized diagnostic components (ClawGuard, cycle detection, LumiMAS, LLM RCA, MIRAGE) are separated from Type I confidentiality claims.

TABLE S5
From prior cross-layer systematizations to result-level deployment decisions: the specific comparison object and its demonstrated use.

Prior work	Established perspective	Added result-level question	Demonstrating analysis
Banerjee et al. [23]	Cross-layer compound AI attacks can relax threat assumptions; MITRE organization	Does cover end at the protected interface, get reserialized, or execute as an action?	NetEcho final-hop check and Dangol/AHB transport/action distinction; explicit conditions for permitting transfer.
Forough et al. [24]	Confidential-computing threat model; partial inference-defense transfer; isolation is not side-channel immunity	Which oracle, execution mode and component-cost result licenses the proposed defense transfer?	Early Bird KV/semantic split and KVGov hardware/simulation/analytic separation.
Cai et al. [25]	Traffic features, shaping objectives and bandwidth trade-offs	Can a traffic-cover cost/utility argument be used for a dummy tool action?	Dangol/AHB: a state-machine boundary determines which task/policy outcomes must additionally be measured.
Debenedetti et al. [26]	Auxiliary ML components induce privacy channels	Distinguish an execution-mediated oracle from ordinary semantic attribution	C3 boundary and mixed-output eligibility judgments.
Lahjouji–Colaco [27]	Agent data surfaces, governance, composition and cross-session risks	Which bridge test would license a specific transfer?	Receiver/intervention and generator/holdout result maps.
Kim; Horváth et al. [28, 29]	Broad agent security; physical DNN mechanisms/threat models	Compare source and receiving privileges without a universal strength order	Remote calibration versus own-site, local and privileged diagnostic profiles.

independence nor family balance is established.

The original 19-fit comparison selects weighted logistic/forest candidates on validation loss only; selected models are forests. H2 adds type bigrams to coarse counts/page-age. Original intervals use 3,000 paired-question resamples and a nominal 98.333% three-contrast allocation, conditional on fixed fitting/selection and noise realizations. Archived H1's identical selected predictor yields exactly zero difference and interval $[0,0]$; it is a selection illustration, not equivalence evidence.

For Q questions with n_q records each, loss is $Q^{-1}\sum_q n_q^{-1}\sum_{i\in q} -\log p_{i,y_i}$; accuracy substitutes the correct-argmax indicator. Fit weights $N/(Qn_q)$ equalize question weight. Add $\varepsilon = 10^{-6}$ to every class probability, renormalize, then clip true-label probability to $[\varepsilon, 1]$. Bootstrap draws Q questions with replacement; original contrasts share draw indices, with percentile endpoints 0.8333% and 99.1667%.

Controls declared with original outcomes known The additive protocol is frozen before new fits, with original outcomes known. All-14 common support is empty. Select the largest identity subset with at least 20 common training questions, then greatest support and lexical tie-break: Gemini 3.1, Gemma 4-31B, Seed 2-lite, with 32/10/10 questions. Hash order selects one row per question/identity, retaining failures; test has 30 rows. This is a three-class sensitivity result.

The first-Wikipedia-document *proxy* stops at HTTP navigation, beforeunload, unmarked origin/URL change or clock reset, but not SPA history alone. Markers gate membership and never enter features. No frame/document IDs authenticate membership; late callbacks may cross boundaries. Empty views remain and epochs are not pooled. The supplement gives the complete six-event projection example and boundary implementation.

Three row-seeded shuffles preserve counts/timestamps/coarse features in *all partitions*, refitting combined candidates with coarse retained. Fits total 8 matched-task, 8 document-proxy, 8 clean and 12 shuffled; all 36 finish before test prediction, without convergence warnings. Clean H2 reproduces exactly. Each shuffle selects coarse: zero increment is then an algebraic consequence of selecting the same predictor. This checks the selection pipeline, not whether shuffled sequence features contain no information; a forced-sequence comparison would be a different, unperformed analysis.

New intervals use 3,000 question resamples, seed 20261018 and 98.333% allocation across the three new contrasts, conditional on fits and shuffle streams. They are exploratory, not blind confirmation or global adjustment across all historical analyses.

TABLE S6
Source–receiver–intervention map. The target column distinguishes content/property inference, attribution and human–agent distinction; none alone establishes an unauthorized privacy violation. U denotes an unresolved source/run binding.

Source / target	Signal origin → receiver	Intervention / evaluated mode	Surviving path or restriction	Comparison consequence
UI Traces / identity attribution	Agent actions → own-site JS	Recorded timestamp delays; offline D2	Types and order preserved	Test residual action features; no live task-cost estimate.
NetEcho / content inference	Serving → network A/B traces	Upstream padding/batching; protected-data fit	Decoded upstream content/state can be reserialized downstream	Inspect final-hop emission; D1 is not automatically D2.
SIGLEAK / skill nondisclosure	Procedure → task/output receiver	Common prompt defense	Procedural signatures remain	Compare same receiver/target; not a paired installation effect.
AHB / human–agent distinction	Touch behavior → app/detector	Executed fake touches; separate transforms	Task state may change; fit binding U	Measure transitions and objective utility; do not import offline delay costs.
WRA / private-intent inference	Real/decoy research requests → visited domains	Concurrent execution; interleaved-domain evaluation	Intent reconstruction falls with one decoy	Preserve domain-view privacy benefit; terminal decoy utility and cost are separately unreported.
Dangol / workflow-property inference	Protocol/traffic → defined link	Stored captures versus live OHTTP (§9.7, OHTTP paragraph after Table 5)	Feature ablation differs from network masking	Keep accuracy 0.17 capture and 0.30 live populations separate.
Early Bird / cache state	Cache reuse → API timing	KV simulation versus GPT-Cache prototype	Distinct matching mechanisms	6ms/4% applies to semantic prototype only.
KVGov / private prompt	Shared cache → probe oracle	Simulated inference; real timing; analytic isolation	Hash-key isolation not semantic-index isolation	Hardware gap does not become deployed defended ASR.
FP-Agent / identity attribution	Policy/browser → visited site	Task/site evaluation, mixed attributes	Static configuration may remain	Task holdout is not independent-visitor holdout.
Traffic benchmark / topology	Multi-agent workflow → selected links	Explicit topology/view shift	Unseen generators differ from hidden links	Cross topology with receiving-view restriction.
LeakyLMs / architecture	Inference → streaming client	Chosen-prompt timing; no measured defense	Same-GPU-class calibration for architecture	API reachability is not hardware-independent recovery.

TABLE S7
Dependency–holdout–claim map. Each row excludes one explanation, not all confounding.

Dependency / source	Required holdout	Claim licensed; residual limit
Task / UI Traces	Question/task-family	New task units; fixed harness remains.
Browser / FP-Agent	Visitor/configuration	Configuration transfer; task-only split does not test it.
Workflow / traffic benchmark	Agent topology or composition	Tested topology shift; no all-orchestrator guarantee.
Hardware / LeakyLMs	GPU/serving calibration	Tested architectural transfer given calibration; not unknown-hardware recovery.
Repository / coding fingerprints	Repository and generation pathway	Tested attribution domain; not covert human-mediated use.
Order / worked case	Same units, refit shuffle	Fixed-program order increment; not task-independent identity.

TABLE S8
Fixed-archive predictive evidence and matched support. Units: nats of question-weighted loss reduction. Positive values favor the second program. Intervals condition on fitted programs; the original and additive-control allocations remain separate.

Contrast	Estimate	98.333% interval
Worked case: coarse vs. combined (H2)	0.3447	[0.2974, 0.3944]
Matched questions, 3 identities[a]	0.2634	[−0.2584, 0.7241]
Reanalysis: clean vs. adapted (H3)[b]	0.1503	[0.1205, 0.1815]

[a] 30 rows/10 test questions, 3 classes, rather than 1,135 rows/226 questions, 14 classes; not a direct effect-size comparison. Its interval uses the additive three-control allocation. Archived identical-predictor H1 remains in the protocol.
[b] Both programs predict the same timestamp-perturbed test rows; three fixed realizations are averaged.
Cross-document order assumes an archive receiver, not verified ordinary-website capture. Intervals vary test-question weights, not training, task families or deployments. No live defense utility or cost is inferred.

Separate code reconstructs points/intervals from saved probabilities. Matched-task uncertainty and the 509/1,135 test error flags (3.4–91.5% by identity) prevent intrinsic-identity claims.

The saved-result verifier supplies local replay; it does not establish independent replication or live utility/cost.

Failure/activity diagnostics The configuration was frozen with prior outcomes known: three representations, four candidates each, original weights, seed 20261008; all 12 fits precede test prediction. Failure-only uses error/absent-result bits; empty-only uses no projected events; activity-only uses log-count plus seven page-age summaries, not cross-document duration. Candidate hyperparameters and projections are supplemental. No new confirmatory intervals are supplied. Flags cover 43.00% overall and 44.85% test rows, not authenticated task success. Stratifying 509 flagged/203 questions versus 626 unflagged/216 questions changes support; increments are not a causal failure effect. Semantic-family balance remains unavailable.

TABLE S9
Public-trace permission projection. "Site" means its own document only; it does not authenticate cross-document capture. Full field-level CSV and one complete ingested-record projection are supplemental.

Field	Site JS	Analysis use
Type, page-local time	Yes	Original predictors; activity diagnostic
Within-document order	Yes	Type bigrams
Cross-document order	Unverified	Archive-receiver bigrams
URL/origin	Own page	Origin gate only; no text/page-key feature
HTTP navigation, collector time	No	Boundary proxy only / time excluded
Error/result metadata	No	Analyst-only failure baseline
Task/answer metadata	No	Excluded; question hash splits/weights only
DOM/link text	Own page	Excluded from all predictors
Document/frame ID	Absent	No authenticated document membership

TABLE S10
Post-review H2 controls. Positive values denote a question-mean log-loss reduction (nats); brackets are conditional 98.333% resampling intervals. Not a new agent run.

Contrast	Test rows / questions	Effect [interval]
Matched tasks, 3 identities: coarse minus sequence	30/10	0.2634 [−0.2584,0.7241]
First-document-epoch *proxy*: coarse minus sequence	1135/226	0.1504 [0.1193,0.1825]
Original clean minus mean of 3 refit shuffle increments	1135/226	0.3447 [0.2961,0.3953]

All 14-class common support is empty. The first-document view is reconstructed, not authenticated. Its 0.1504 feature increment and H3's 0.1503 timestamp-adaptation effect are distinct contrasts on the same archive, not independent replications. Cross-document order assumes an archive receiver. All three shuffles select coarse, so their zero increments make this contrast's 0.3447 point equal to original H2. The intervals use separate frozen bootstrap seeds (20261018 here; 20261008 for original H2), yielding different finite-resampling endpoints. Zero selected increments do not establish information equivalence. No task utility or live defense is measured.

Exact H3 perturbation and adaptation The unchanged perturbation sets $t'_{ire} = t_{ie} + u_{ire},\ u_{ire} \sim U(0, 5000\,\text{ms})$, independently per valid page-local timestamp in source-array order. Missing times consume no draw; types/order/page keys remain fixed. Three row-seeded realizations transform all partitions. Each adaptation clones the clean-selected representation/architecture and fits once on perturbed training rows with identical weights, without retuning or additional data (three fits). Both programs predict the same perturbed test rows; average paired losses across realizations, rows within question, then questions. The supplemental exact protocol supplies entry points, seed strings and invariants. The transformation edits stored times; it does not execute action delays.

S5 Intervention Inventory and Evidence Dispositions

The companion mode-specific table and 33-record ledger distinguish executed inference, offline/simulated evidence and component costs. All 66 utility/cost positions retain the source, scope, state and consequence for a named claim or inventory-only role. The main deployment-question table does not treat these heterogeneous records as a common success/failure denominator.

TABLE S11
Additive failure/activity diagnostics on all 1,135 test rows/226 questions. Point scores use the original question weights; no new confirmatory intervals. Original programs are read from saved probabilities, not refitted.

Program	Loss (nats)	Accuracy (%)
Training prior (saved)	2.64267	6.59
Error/result flags only[a]	2.47464	14.25
Projected-empty flag only	2.63832	6.79
Activity/page-age only[b]	1.81732	37.68
Original coarse (saved)	1.44058	51.29
Original coarse+sequence (saved)	1.09591	66.26

[a] Analyst-only diagnostic, not an observer-valid attacker; two binary flags, not error text. [b] Total event count and seven local-page-time summaries; no action types/order, diagnostics, text or session duration. Same four candidate architectures and validation-only selection for each of the three new representations (12 fits). Low-dimensional baselines do not rule out higher-order failure/task confounding.

TABLE S12
Quantitative assertions excluded for missing bindings.

Source	Missing binding for quantitative use
AgentPrint	Referenced supplementary mitigation result, procedure and population not located.
Weiss	No matched vendor-defense before/after outcome located; the artifact appendix reproduces an attack table.
Gu	No provider-change result with a matching comparison population located.
PromptEcho	MSS/threshold padding is described; exact threshold, population, cost denominator and defended outcome are unbound.

Sources: [30, 31, 32, 33]. No missing value is imputed; withholding these assertions does not remove the four families.

Four textual defense assertions lack the quantitative binding needed for a matched effect (Table S12). Their unbound quantities are excluded from comparison; their qualifying attack evidence remains. AHB has verified condition-wise refit code, but historical D1-run and paired D2 are recorded as NR-V, not reconstructed. Its task outcome is not inferred from that code.

D1 denotes protected-condition fitting/calibration; D2 denotes a paired training-procedure comparison on the same defended test condition. Neither entails feasible labeled-data access or matched budgets. Token payload, plaintext padding, wire bytes, model loading, map lookup and added service latency remain distinct cost objects. A scoped NR-R completes the requested extraction, rather than asserting zero cost or preserved utility.

S6 Conclusion Sensitivity

Table S14 exposes direct-source dependence in the original snapshot. Within that snapshot, removing NetEcho withdraws the only downstream re-emission instance; removing AHB withdraws the measured mobile state-change instance, while WRA's privacy-oriented decoy result remains. Removing WRA withdraws that private-intent decoy result while AHB retains the task-state branch. UI Traces and SIGLEAK remain different surviving-feature instances, not replications. Removing Dangol withdraws its capture/live contrast; Early Bird, KVGov

TABLE S13
Scoped intervention dispositions across all 58 core families in the full collection (alongside the 43-core October 8 snapshot and 15 new core additions). Categories are mutually exclusive at the family level.

Source-scoped disposition	Snapshot (43)	Added (+15)	Full (58)
Evaluated mitigation component/condition	16	+4	20
Fixed common guardrail (not paired installation)	2	+0	2
Other intervention / decoy boundary / objective	4	+1	5
No evaluation identified in inspected scope	13	+9	22
Type IV diagnostic component (confidentiality NA)	4	+1	5
Unbound assertions excluded from effect comparison	4	+0	4
Total core families (Result records)	43 (33)	+15 (+6)	58 (39)

The four October 21 evaluated-mitigation additions (`wu2026cacheprune`, `lee2026otro`, `chen2026sparleak`, `hadad2026shape`) contribute six source-bound result records (bringing the result ledger from 33 to 39 records across 78 utility/cost fields); `politis2026splitdecoy` evaluates forward split-model decoys against backward gradient-support leakage, and `revankar2026mirage` is the fifth Type IV diagnostic component.

and SpliceDefense retain separate component examples. S9/S10 separately report the later controlled evidence and source expansion; they are not part of this historical ablation denominator.

Starting from the original 60/43/17 total/core/adjacent families, deleting AHB, WildAuth and coding fingerprints leaves 57/42/15; deleting its four diagnostics leaves 56/39/17; joint removal leaves 53/38/15. These are snapshot ablations, not the later 82-family collection. The snapshot's four diagnostic/semantic-boundary additions are not premises of the cover/action contrast. For its five multifeature-trace studies, unresolved bindings bound D1 at three–five and D2 at two–five; removing AHB gives three–four and two–four. Early Bird calibration is a separate stage.

Excluding all four unbound defense quantities leaves the three deployment cases unchanged, because none uses those quantities as a premise. AHB's detector-run unknowns prevent an adaptive-detector claim but do not erase its separately reported task-success points. Its non-Fake online point is an additional Table 1 subcondition in the case ledger, not an extra independent experiment or an imputed value in the combined-condition inventory row. Whole-service costs, training access and missing utility admit no favorable imputation or ranking.

Narrowing H2 to matched tasks leaves task-independent identity unresolved; the document proxy does not authenticate capture. H3 supports only same-record offline adaptation. The original joint workflow endpoint remains negative, including the missing-score bounds and post-outcome sensitivity views. These positive, inconclusive, zero and negative outcomes are preserved with their original endpoint meanings.

S7 RETAINED WORKFLOW COMPARISON AND RETROSPECTIVE VALIDITY BENCHMARK

S7.1 Diagnostic decomposition of the retained 512-root synthetic comparison

The historical model-assisted task compares a flat checklist (`R`), active structured checklist (`T`) and raw SCOPE receiver traversal (`V`) across 512 primary roots in `gpt-6-luna`/low sessions. On the strict *joint* string-matching endpoint (requiring the binary decision, exact configuration-count strings, evidence identifiers and JSON schema all to match simultaneously), correctness is 40.63% (`R`), 45.12–45.31% (`T`) and 32.16% (`V`), with conditional 95% Hoeffding envelope $[-25.16, -0.95]$pp for `V` − `T` ($h = \sqrt{2\log(2/0.05)/512} = 0.12004035$).

Decomposing the errors explains why: `V` produced 695 decision-right/count-wrong judgments versus `T`'s 476, with zero scorable schema or evidence-ID failures. Evaluated on the **binary deployment decision itself**, `V` outperformed `T` by +1.11 to +1.30 percentage points. We retain the negative joint-endpoint result in full and therefore use SCOPE strictly as an indexing schema, while executing deployment audits via the three-check procedure evaluated below.

S7.2 24-scenario retrospective deployment-transfer benchmark (M1 validity study)

To test whether applying the three-check result-record procedure improves deployment transfer decisions over headline-metric reuse on real literature cases, we evaluated a 24-scenario retrospective benchmark (`experiments/revision-m1-m8/m1_validity_benchmark.csv`; Main Paper Table V). The benchmark comprises 12 documented deployment failures/mismatches (*Incompatible*), 5 missing-bridge cases (*Unresolved*), and 7 valid source-local reuses (*Compatible*), evaluated by a read-only statistical verifier (`evaluate_coding_and_validity.py`).

An objective, deterministic headline-metric baseline rule—which predicts *Compatible* whenever the source paper reports local attack-accuracy reduction or $< 10\%$ component overhead (true of all 24 published defense scenarios)—achieves **29.2% (7/24)** overall accuracy and **0.0% (0/12)** recall on the 12 documented deployment failures. Two independent expert coders applying the three-check record procedure achieve **95.8% (23/24)** and **91.7% (22/24)** accuracy (mean **93.8%**, +64.6pp over baseline; failure-detection recall **100.0%** and **91.7%**), with **87.5% (21/24)** raw inter-coder agreement and **Cohen's** $\kappa = 0.795$ (95% bootstrap CI $[0.538, 1.000]$).

S8 RESULT-PAIR COMPARABILITY, NINE-ITEM SPLIT AUDIT AND DUAL-CODING

This audit applies the paper's comparison rules to the 33 result records across 22 families (and the nine-item split reporting checklist to the 16 baseline and 4 newly added evaluated-mitigation families). In addition to deterministic rule aggregation, a domain-expert human annotator independently dual-coded the audit cells and pairwise judgments (`experiments/revision-m1-m8/`).

TABLE S14
Load-bearing source ablation within the original 60-family snapshot. Entries identify its actual dependence; S9/S10 report subsequent controlled evidence and source expansion separately.

Conclusion / source locator	What is supported	What is lost on source removal / NR dependence
Action preservation / UI Traces 6.1,Fig5	Recorded delays preserve structure and allow adaptation	Action-delay instance and one D1/D2 positive lost; exact release-point binding not needed for protocol claim.
Procedural preservation / SIGLEAK 4/C	Skill inference persists under common extraction defense	Skill-defense instance lost; no paired installation-effect claim.
Re-emission / NetEcho IV-C–E	Downstream reserialization can restore fine granularity	Only direct downstream re-emission instance lost; no multisource replication claimed.
State change / AHB 5.2.3/Table1/E.3	Fake actions can cause unintended UI transitions	Only executed mobile state-change instance lost; detector-run binding remains U.
Private-intent cover / WRA 8.1/Table8	Real/decoy execution with interleaved-domain inference evaluation lowers one-decoy E_{func}	Private-intent decoy instance lost; AHB state-effect instance remains. Same-condition terminal utility/cost are NR-R.
Mode split / Dangol §9.7, OHTTP paragraph after Table 5; 9.8–9.9/Table6	Capture frontier differs from live protocol and timing ablation	Concrete accuracy 0.17/0.30 pair and one D1/D2 positive lost.
Cost boundary / Early Bird V-A/B; Splice VII-C; Speculation 6.1; KVGov 5.9	Separate component and deployment costs	Removing one withdraws its example, not other boundary instances; full service costs remain unknown.
Generator dependence / UI Traces, FP-Agent, traffic benchmark, LeakyLMs	Task, browser, topology and calibration tests differ	Each source's axis withdrawn when removed; no pooled portability estimate.

TABLE S15
Descriptive comparability within the 13 same-family pairs and 102 related-cohort pairs. Independent expert dual-coding over 50 pairs (150 judgments) reached 94.0% agreement ($\kappa = 0.916$).

Subset	Compatible	Incompatible	Unresolved
Same family ($n = 13$)	7 (53.8%)	6 (46.2%)	0
Related cohort ($n = 102$)	16 (15.7%)	80 (78.4%)	6 (5.9%)

S8.1 Within-family and within-cohort pairwise comparability

While the machine-readable inventory retains all $\binom{33}{2} = 528$ unordered pairs for completeness, 426 pairs cross unrelated mechanism cohorts and are outside scope by definition. Table S15 summarizes the two informative subsets: the **13 within-family pairs** and the **102 within-cohort related pairs** (traffic-envelope: 66; cache-service: 28; behavioral humanization: 3; generator fingerprint: 3; local-memory: 1; fixed guardrail: 1).

Within the 13 same-family pairs, seven compare aligned outcomes and six substitute different primitives or cost objects. Of the seven aligned-outcome pairs, four have positively different collection or execution populations (such as Dangol's stored-capture feature ablation at 0.17 accuracy versus live loopback OHTTP at 0.30 accuracy) and three lack a verified paired-test binding; none licenses direct cross-record subtraction. Across the 50 dual-coded pairs (all 13 same-family pairs plus 37 stratified related-cohort pairs = 150 judgments across descriptive, paired-effect and transfer rules), independent two-coder agreement is **94.0% (141/150)** with **Cohen's** $\kappa = 0.916$ (95% bootstrap CI $[0.856, 0.963]$; `m7_dual_coding_pairs.csv`).

S8.2 Nine-item split reporting audit and independent dual-coding

Table S16 splits Check 4 into **4a (D_1 protected-condition fitting/calibration)** and **4b (D_2 matched clean-versus-adapted test comparison)**, and splits Check 6 into **6a (defended inference outcome)**, **6b (task or component utility)** and **6c (mechanism or protocol cost)** across both the 16 original evaluated-mitigation families (144 cells) and the 20 total evaluated-mitigation families (including `wu2026cacheprune`, `lee2026otro`, `chen2026sparleak` and `hadad2026shape`; 180 cells).

TABLE S16
Nine-item split reporting audit over the 16 original evaluated-mitigation families (and 20 total including the four October 21 mitigation additions in parentheses). Splitting Check 4 (D1 vs. D2) and Check 6 (inference, utility, cost) exposes component-level variance without requiring every study to run an end-to-end deployment evaluation.

Audit item	Bound	Partial	NR-R	NR-V	NA
1. Target & protection polarity	16 (20)	0 (0)	0 (0)	0 (0)	0 (0)
2. Receiver fields & linkage	13 (17)	3 (3)	0 (0)	0 (0)	0 (0)
3. Placement & action dispatch	13 (16)	2 (3)	0 (0)	1 (1)	0 (0)
4a. Protected-condition fit (D1)	4 (4)	1 (1)	0 (0)	4 (6)	7 (9)
4b. Matched adapted test (D2)	2 (2)	2 (2)	0 (0)	5 (7)	7 (9)
5. Workloads & generator shifts	14 (18)	2 (2)	0 (0)	0 (0)	0 (0)
6a. Defended inference effect	14 (17)	0 (1)	2 (2)	0 (0)	0 (0)
6b. Task or component utility	2 (3)	2 (3)	10 (11)	0 (0)	2 (3)
6c. Mechanism or protocol cost	10 (11)	1 (3)	4 (5)	0 (0)	1 (1)

Bound: explicitly established for the scoped claim. Partial: a supported element coexists with a proxy or unlinked operating point (e.g., AHB D1-code without verified Table 1 D1-run; WRA Table 11 visibility utility separate from Tables 8–9 decoys). Only one family across the expanded 20-family mitigation set (`wu2026cacheprune`) jointly binds 6a, 6b and 6c on the same operating point. Independent two-coder agreement over the 144 baseline cells is 93.1% ($\kappa = 0.876$, 95% CI $[0.798, 0.943]$).

Across the 144 baseline cells, independent expert dual-coding (`m7_dual_coding_checks.csv`) reached **93.1% raw agreement (134/144)** and **Cohen's** $\kappa = 0.876$ (95% bootstrap CI $[0.798, 0.943]$). All 10 pre-adjudication differences occurred across adjacent *Partial* versus *Bound*/NR-R/NR-V boundaries (e.g., whether AHB's verified refit code without Table 1 run provenance is *Partial* or NR-V on Check 4a, and whether WRA's separate Table 11 visibility utility is *Partial* or NR-R on Check 6b). Full cell-by-cell rationales and source locators are preserved in `m6_split_checks.csv` and `m7_dual_coding_checks.csv`.

S9 Controlled Boundary Tests: Real TLS 1.2 Streams and Stateful Tools

The controlled experimental suite combines the 72 frozen `gpt-6-luna` outputs/plans (48 topic explanations for E1/E3 and 24 tool plans for E2) with real loopback TLS 1.2 socket replays and oracle-free retail cover evaluations (`experiments/revision-m1-m8/run_sp2027_experiments.py`).

Check 1 & Check 3 on real loopback TLS 1.2 sockets (E1, E1-Grad & E3) Every explanation traverses real loopback TCP sockets wrapped with TLS 1.2 (`ECDHE-RSA-AES128-GCM-SHA256`) through a transparent byte-capturing relay that parses TLS record type 23 application frames and verifies exact UTF-8 plaintext SHA-256 recovery at the receiver. Across the four natural inference regimes on the saved corpus (Main Paper Table VIII, Panel B):

- T_1 **(Directive verbosity: 1-sentence brief vs. 6–8 sentence full response,** $d = 10.4$**):** Unpadded (721B, 1.00×), 256B-bucketed (857B, 1.19×) and 512B-bucketed (1,086B, 1.51×) terminal streams remain 100.0% separable; 1,024B bucketing (1,401B, 1.94×) reduces D_2 accuracy to **70.8%** (95% CI $[58.3\%, 83.3\%]$), and constant 16-frame capping (16,848B, 23.37×) reaches **50.0%**.

- T_2 **(Short vs. medium exposition: 1-sentence vs. 2-sentence complete sentences,** $d = 3.1$**) and** T_3 **(Medium vs. extended exposition: 2-sentence vs. 3-sentence complete sentences,** $d = 1.4$**):** Unpadded accuracy is 100.0% (T_2) and 91.7% (T_3, CI $[79.2\%, 100\%]$); 256B bucketing reduces accuracy to **87.5%** (T_2, CI $[75.0\%, 100\%]$) and **62.5%** (T_3, CI $[50.0\%, 75.0\%]$); and 512B bucketing (628B and 654B mean wire bytes, 0.87× and 0.91× baseline) reduces both T_2 and T_3 to **50.0%**.

- T_4 **(Engineering vs. ecological domain on full unmodified responses,** $d = 0.25$**):** Unpadded test accuracy is **45.8%** (CI $[25.0\%, 66.7\%]$), confirming that size-only leakage vanishes when target classes share matched length distributions.

In Experiment E3 (Main Paper Table VIII, Panel C), comparing a clean-trained attacker (D_0, threshold 623B) against defended-trained attackers (D_2, trained on $k = 4$ or $k = 12$ topic pairs, plus a 3-feature regularized linear classifier over TLS record count, total payload bytes and final-record remainder) on the identical defended test streams yields:

- **Bucketed 1024B quantum (**1.94×**):** D_0 scores **50.0%**, whereas D_2 ($k = 4$ or 12) rebounds to **70.8%** ($\Delta = +20.8$pp, CI $[+8.3, +33.3]$).

- **Stochastic chunk-tail padding** $+U(0, 512\mathbf{B})$ **(**1.44×**) and** $+U(0, 1024\mathbf{B})$ **(**1.88×**):** Under $+U(0, 512\text{B})$, D_0 scores **86.4%** while D_2 rebounds to **99.9%** ($\Delta = +13.5$pp, CI $[+10.7, +16.3]$). Under $+U(0, 1024\text{B})$, D_0 scores **68.2%**, whereas D_2 rebounds to **90.2%** ($k = 4$), **89.3%** ($k = 12$, $\Delta = +21.1$pp, CI $[+17.3, +24.8]$) and **89.9%** (3-feature linear D_2).

- **Deterministic offset (**+512**B,** 1.87×**) and Constant cap (**16,848**B,** 23.37×**):** +512B offset rebounds from **50.0%** (D_0) to **100.0%** (D_2, $\Delta = +50.0$pp); constant capping remains at **50.0%** for both ($\Delta = 0.0$pp).

S9.1 Retail execution bridge and oracle-free realistic cover policies: joint privacy and utility

The retail execution study uses Sierra's pinned τ-bench repository (commit `59a200c6d575d595120f1cb70fea53cef0632f6b`; 16 retail tools, 115 public test reference tasks, 3 seeds, `before` vs. `after` insertion). Because seeds and positions are repeated measures on the same 115 tasks, the effective sample size is **115 tasks**: **97 eligible tasks** (where the user owns ≥ 1 initially `pending` order = 291 task-seed units per position) and **18 ineligible tasks** (zero pending orders = 54 units per position where no cover executes and reward is mechanically 100.0%).

Exact decomposition of before versus after cover call success (M3 & M4) From `m3_m4_exact_breakdown.json` (Main Paper Table IX, Panel C):

- **Pre-insertion (`before`, 97 eligible tasks / 291 units):** The target order is initially `"pending"` in all 291 eligible units, so `cancel_pending_order` **succeeds and mutates state in 291/291 (100.0%) of eligible units**. Native reward drops to 0.0 in 258/291 (88.66%) units and remains 1.0 in **33/291 (11.34%) units** (CI $[5.84\%, 17.87\%]$; error-free eligible $n = 82$: **9.76%**). In all 33 units (12 tasks: IDs `16, 30, 31, 32, 38, 54, 55, 59, 66, 69, 81, 99`) where `native_reward` remained 1.0, the reference program itself called `cancel_pending_order` on that exact order with reason `"no longer needed"`; pre-insertion cover cancelled it first, causing the reference program's subsequent call to return `Error: non-pending order cannot be cancelled` while leaving the final database snapshot identical to baseline.

- **Post-insertion (`after`, 97 eligible tasks / 291 units):** During prior reference execution, the reference program modifies the selected order's items (`modify_pending_order_items`, setting `status="pending (item modified)"` in 79 units) or cancels the order (`cancel_pending_order`, setting `status="cancelled"` in 41 units) in **120/291 (41.24%) of eligible units**. Consequently, the post-insertion cover call is **rejected with `Error: non-pending order cannot be cancelled` in 120/291 (41.24%) of eligible units** (leaving state untouched and reward at 100.0% in all 120 units) and **succeeds in only 171/291 (58.76%) of eligible units** (where reward drops to 0.0% in all 171 units), yielding the **41.24%** eligible post-insertion reward (CI $[32.30\%, 50.17\%]$; error-free eligible $n = 82$: **38.21%**).

Oracle-free realistic cover policies and D_0-vs-D_2 privacy evaluation File `m2_retail_privacy_utility.json` (Main Paper Table IX, Panel B) jointly evaluates task

utility and observer cancellation-intent inference (D_0 vs. D_2 3-feature linear classifier over observed cancellation verb, write count and write response bytes across the 97 eligible tasks, split 48 train / 49 test):

1. `live_idempotent_address`: Overwrites the pending order's shipping address with its current values; preserves **100.0% native reward** (`before` and `after`), while a cancellation-verb observer retains **100.0%** accuracy (D_0 and D_2).

2. `live_idempotent_item_swap`: Maps the first item to its own `item_id` (Δprice $= 0$); mutates state in **59.45%** (`before`) and **32.30%** (`after`) of eligible units by setting `status="pending (item modified)"` and appending a $0.00 refund record, dropping eligible native reward to **40.55%** (`before`) and **67.70%** (`after`) while leaving verb-observer accuracy at **100.0%**.

3. `live_instruction_aware` (**strictly oracle-free**): Parses *only* `task.instruction` and initial order items (never reading `task.actions`) to select an unmentioned pending order when available (70.10% [68/97] of eligible tasks) and falls back to `modify_pending_order_address` otherwise. It raises task-touched state equality (`gold_touched_state_equal`) from 62.54% to **78.35% (`before`) and 89.69% (`after`)**, while whole-account `native_reward` is **35.05% (`before`) and 46.39% (`after`)**. On the privacy axis, it drops clean-trained D_0 observer accuracy from **100.0% to 51.02%**, whereas a defended-trained D_2 observer rebounds to **73.47% (`before`) and 71.43% (`after`)**.

4. `live_decoy_account`: Cancels a pending order on a designated decoy user account, preserving **100.0% of the active user's account state (`user_orders_state_equal = 100.0%`)** and **100.0% of task-touched state**, while reducing D_0 observer accuracy to **22.45%** (D_2 rebounds to **71.43%** by exploiting session write counts).

S10 Completed Finite Retrieval Update and Exact Reconciliation

Four Semantic Scholar bulk queries (`S1R`, `S2`, `S3`, `S4`) and two native arXiv searches (`ARXIV-A1`, `ARXIV-A2`) were completed on October 21, 2026, returning 615 occurrences deduplicated to 565 unique records. Combined with the original 60-family October 8 snapshot, the final merged index contains 82 families: 58 core (20 workflow-origin, 38 inherited; 53 non-diagnostic and 5 Type IV diagnostic) and 24 adjacent.

S10.1 Executable queries and pre/post-audit disposition accounting

The Semantic Scholar bulk queries (2023–2026) and native arXiv CS searches (January 1, 2023–October 21, 2026) are:

S1R (116): (agent | workflow) + (LLM | "language model") + ("side channel" | "side-channel" | fingerprinting).
S2 (120): (LLM | "language model") + (traffic | streaming) + (leakage | side-channel | fingerprinting).
S3 (15): (LLM | "language model") + (padding | "re-emission") + (privacy | leakage | side-channel).
S4 (287): (agent | LLM) + (decoy | "cover traffic" | "fake actions" | "cover actions").
ARXIV-A1 (50): `"side-channel" AND "large language"`; ARXIV-A2 (27): `"side-channel" AND agent`.

Table S17 provides the complete item-by-item reconciliation before and after the 204-record exclusion audit (177 high-risk pattern matches plus 27 reason-stratified samples, seed `20261021`). Before the exclusion audit, screening identified 221 C2/C3 exclusions (`EXCLUDE_NO_HIDDEN_AUXILIARY_TARGET`), 14 new core families, 23 existing core families, and 28 source-reviewed adjacent contexts. The exclusion audit rerouted exactly four records out of the 221 C2/C3 exclusions ($221 - 4 = 217$):

1. `SUP-0192` (MIRAGE [34]): rerouted to `NEW_CORE_SOURCE_VERIFIED` ($14 + 1 = 15$ final new core families, all 15 incorporated into the 82-family index via `addition.csv`).

2. `SUP-0552` (InputSnatch arXiv version [35]): rerouted to `EXISTING_CORE` ($23 + 1 = 24$ existing core recoveries; merged as an alias of the already-indexed snapshot family `inputsnatch2024`, adding $+0$ net new families).

3. `SUP-0191` (LLM Self-Recognition) and `SUP-0424` (Reverse-Engineering Model Editing): rerouted to `ADJACENT_SOURCE_REVIEWED` ($28 + 2 = 30$ final source-reviewed adjacent contexts). Of these 30 source-reviewed adjacent contexts, 7 boundary-defining studies (`addition-adjacent.csv`) are incorporated into the 82-family index ($17 + 7 = 24$ adjacent), while the other 23 (including `SUP-0191` and `SUP-0424`) remain in the 565-record disposition ledger as non-indexed context.

Thus the family transition is strictly additive and self-consistent: 60 (snapshot) + 15 (14 pre-audit + 1 audited new core) + 7 (selected adjacent) = 82 families ($43 + 15 = 58$ core; $17 + 7 = 24$ adjacent).

TABLE S17
Post-audit disposition flow for the completed 565-record supplementary frame (see Main Paper Table 2 for pre-audit vs. post-audit Δ).

Stage or final post-audit route	Count
Complete-query occurrences / duplicates merged	615 / 50
Unique records with an explicit route	565
Excluded: no hidden auxiliary target ($221 - 4$ rerouted)	217
Excluded: outside LLM execution / different target	156 / 19
Existing core ($23 + 1$ alias) / adjacent recovered	24 / 1
New source-bound core families ($14 + 1$ MIRAGE; all indexed)	15
Adjacent source-reviewed ($28 + 2$; 7 indexed) / metadata-only	30 / 28
Other security background / reviewed background / proposal	53 / 4 / 1
Missing abstract / unavailable primary full text	16 / 1

S10.2 Seed overlap clarification and dual-coding reliability

As noted in Main Paper Section 2, the 24 fixed known papers (`known24`) are a 24-paper core subset of our 40 initial citation-tracing seeds (`seed_n=24` in `check_known24_offline.py`), not an external holdout. Their 17/24 query recovery (and 25/60 snapshot recovery) measures lexical overlap between the six Boolean keyword queries and our seed-and-snowball corpus. Independent human/expert dual-coding of a 100-record stratified subsample of the 565-record frame (`m7_dual_coding_screening.csv`) achieved 93.0% three-class agreement ($\kappa = 0.886$, 95% CI $[0.801, 0.963]$) and 95.0% binary Core-inclusion agreement ($\kappa = 0.880$, CI $[0.764, 0.975]$).

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